

GradeSense

Explainable AI Decision Support for Paper Grade Transitions

Technical Report

Honeywell Hackathon Final Submission | 26 July 2026

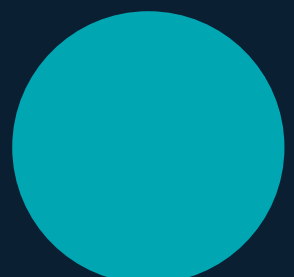
Scope

Early basis-weight risk prediction, direct multi-horizon forecasting, explainable recommendations, operator decision capture, and production deployment.

Repository: github.com/bhavukm007/GradeSense

Live deployment: gradesense-4weh.onrender.com

Demo: drive.google.com/file/d/1rCr3CzuCit0O029VzyxIhHoR1lv9E93E/view



Key Results and Solution Snapshot

Performance and implementation evidence summarized from the final presentation

0.920

Crossing ROC-AUC

1.388

Trajectory MAE

0.962

Stabilization R²

2,544

Held-out windows

Governed sensor-to-action workflow

01

SENSE

Live process data

02

PIPELINE

Validate + engineer

03

PREDICT

Snapshot + forecast

04

EXPLAIN

Drivers + confidence

05

RECOMMEND

Constrain + simulate

06

DASHBOARD

Decide + learn

Technical feasibility

- Cloud deployed with API-driven FastAPI and React modules
- Real-time WebSocket monitoring and predictive analytics
- Dockerized services with PostgreSQL persistence
- Versioned model registry, audit history and explainable AI

Business relevance

- **Earlier intervention before quality moves outside specification**
- **Faster operator decisions through explained, ranked actions**
- **Lower rejection opportunity and improved process stability**
- **Human-in-the-loop control with traceable decision history**

Candidate

Bhavuk Mahajan | Candidate ID: 20484629 | Category: Software

Executive Summary

GradeSense is an industrial decision-support platform designed for paper-grade transitions, where interacting process variables can push Basis Weight outside the target specification before conventional monitoring supports an effective response. The system predicts risk, forecasts the future trajectory, explains contributing variables, evaluates constraint-safe interventions, and records the operator decision and observed result.

95.18%	95.72%	92.03%	89.72%
Snapshot quality-score R2	Off-spec probability R2	Crossing ROC-AUC	Crossing F1 at 0.50

Evaluation boundary. The reported values come from the supplied immutable model artifacts and a reproducible evaluation of the stored crossing classifier on the same held-out transition split used during training. Data is synthetic; these figures demonstrate technical feasibility and must not be represented as mill acceptance results.

Report Map

Section	Evidence
Problem and objectives	Challenge mapping, operational constraints, business relevance
Architecture and flow	Frontend, API, models, database, WebSocket, Render
Machine learning	Features, split methodology, regression metrics, classification metrics
Recommendation and XAI	Constraint checks, simulation, ranking, local sensitivity
Results and limitations	Measured artifact performance, honest production-readiness boundary

1. Problem Statement and Objectives

During automatic grade or recipe transitions, Basis Weight can deviate more than 2.5% from its target while machine speed, stock flow, steam pressure, moisture, caliper, pulp consistency, refining energy, headbox pressure, reel tension, ambient conditions, and grade identity interact over time.

Current industrial challenge

- Risk is often recognized after the trajectory has already moved toward an off-spec condition.
- Historian, alarm, QCS, recipe, operator-action, and outcome evidence is fragmented.
- A probability without trajectory, cause, confidence, and safe action is insufficient for operations.
- Recommendations must obey equipment and recipe constraints and remain advisory.

Hackathon objectives

Objective	GradeSense response
Predict off-spec risk	Snapshot risk plus direct 12-step Basis Weight trajectory
Estimate time to deviation	First predicted crossing step and remaining safe time
Recommend safe setpoints	Constraint validation, reforecasting, and improvement ranking
Reduce stabilization time	Compare predicted stabilization before and after intervention
Explain decisions	Feature sensitivity, inference sources, trajectory effects, uncertainty
Capture human response	Accept, reject, modify, delay, apply, expire, and evaluate lifecycle

2. Proposed Solution

GradeSense combines two complementary model families. The snapshot model summarizes the current transition state. The sequential model uses ordered history to forecast the future Basis Weight path and crossing probability. A recommendation engine simulates only feasible interventions through the active forecast model; it never writes to plant control equipment.

Layer	Responsibility	Primary technology
Operator workspace	Live status, prediction, recommendations, alerts, analytics history	React, TypeScript, Tailwind, Recharts
Application API	Validation, orchestration, persistence, governance	FastAPI, Pydantic v2, SQLAlchemy 2
Intelligence	Snapshot prediction, forecasting, explainability, recommendation	scikit-learn, pandas, NumPy, joblib
Streaming	Sensor and prediction events	WebSocket broadcaster with bounded buffers
Persistence	Predictions, alerts, decisions, outcomes, registry, audit	PostgreSQL / SQLite
Deployment	Static frontend and API service	Render, Docker, Uvicorn, Nginx

Human-in-the-loop safety principle

The engine ranks advisory actions. It does not implement closed-loop control. Operators remain accountable for acceptance and application, while the platform preserves the evidence trail needed to evaluate whether guidance was useful.

3. System Architecture

PROCESS / OPERATOR	Sensor snapshots, transition identity, operator decisions
↓	
REACT WORKSPACE	Dashboard, prediction, recommendation, alerts, analytics, history
↓ HTTP + WebSocket	
FASTAPI SERVICE	Schema validation, orchestration, audit, model governance
↓	
INTELLIGENCE SERVICES	Snapshot model forecast model explainability recommendation
↓	
DATA + ARTIFACTS	PostgreSQL/SQLite immutable joblib models checksums

Architectural properties

- Inference loads pre-trained artifacts read-only; startup does not train or regenerate data.
- Model registry promotion validates SHA-256, version, feature schema, metrics, and required interfaces.
- Prediction, recommendation, operator action, outcome, alerts, and audit records are persisted.
- The WebSocket broadcaster supports real-time UX while REST endpoints preserve deterministic workflows.

4. End-to-End System Flow

Step	Processing	Output / control
1. Ingest	Validate current grade, target grade, and process values	Typed ProcessInput; invalid values rejected
2. Snapshot	Preprocess numeric/categorical features and run multi-output Random Forests	Quality score, risk probability, stabilization estimate
3. Forecast	Engineer temporal windows and run horizon-specific regressors/classifiers	12-step median path, residual bands, crossing probability
4. Explain	Replace one variable with learned baseline and measure prediction delta	Ranked local sensitivities; non-causal
5. Recommend	Enumerate constraint-valid adjustments and reforecast each candidate	Ranked feasible actions; unsafe/non-improving omitted
6. Decide	Operator accepts, rejects, modifies, delays, or applies	Persistent decision and audit entry
7. Evaluate	Compare aligned future observations with baseline/intervention	Accuracy, crossing, deviation, stabilization outcomes

WebSocket event flow

The streaming worker emits sensor updates, predictions, drift, alerts, and recommendations to connected dashboards. Queues and recent-sample buffers are bounded. The current implementation is intentionally single-process; horizontal scale requires a shared broker and metrics backend.

5. Data and Validation Methodology

20,000	57,600	240	2,544
Snapshot records	Sequential samples	Transitions	Validation windows

Dataset / model	Training and validation design
Snapshot model	20,000 synthetic transition records; artifact persists validation MAE and R2 for three continuous targets.
Sequential forecast	20-sample history, 12-step horizon, stride 4. Windows never cross transition boundaries.
Split	Entire transitions shuffled with seed 42; 192 transitions for training and 48 for validation.
Classifier evaluation	Stored classifier evaluated on 2,544 held-out windows at threshold 0.50.
Uncertainty	Per-horizon 90th-percentile absolute validation residual used as empirical interval width.

Why transition-level splitting matters

A row-level split could leak nearby samples from the same physical transition into both training and validation. Grouping by transition prevents that leakage and is a more credible estimate of behavior on an unseen transition trajectory.

Metric interpretation

- **R2** describes variance explained for continuous targets; it is not classification accuracy.
- **MAE** is the mean absolute prediction error in the target's native unit.
- **ROC-AUC** measures ranking ability across thresholds.
- **Precision, recall, F1, and accuracy** below use a fixed crossing threshold of 0.50.

6. Snapshot Prediction Model

The snapshot artifact applies median imputation and scaling to numeric inputs, mode imputation and one-hot encoding to grade identity, and a multi-output Random Forest regressor to predict quality score, off-spec probability, and stabilization time.

Target	MAE	R2	Interpretation
Quality score	2.6694	0.9518	Explains 95.18% of validation variance
Off-spec probability	0.0423	0.9572	Mean absolute probability error 4.23 percentage points
Stabilization time	4.4483	0.9622	Explains 96.22% of validation variance

95.18%	95.72%	96.22%
Quality-score R2	Off-spec-probability R2	Stabilization-time R2

Important: classification precision and recall are not applicable to this artifact because its off-spec output is trained and evaluated as a continuous probability regression target. Reporting 'accuracy' for this regression would be technically incorrect.

Artifact provenance

Version: rf-20260725132623266254-7709892b | Training records: 20,000 | Trained: 25 July 2026 UTC | Dataset checksum retained in the artifact.

7. Sequential Forecast Model

The forecast artifact uses direct multi-horizon prediction: one HistGradientBoostingRegressor per future step and a GradientBoostingClassifier for any specification crossing within the supported horizon. Direct horizons avoid recursively feeding predictions back as inputs.

1.3883	0.9203	10,176	2,544
Mean trajectory MAE	Crossing ROC-AUC	Training windows	Validation windows

Model component	Configuration / evidence
History and horizon	20 ordered samples -> 12 directly predicted future steps; stride 4
Regressors	HistGradientBoostingRegressor; horizon-specific models
Crossing classifier	GradientBoostingClassifier; predicts any >2.5% deviation within horizon
Confidence bands	Median forecast +/- per-horizon 90th-percentile absolute validation residual
Validation separation	48 complete transitions held out from the 192-transition training set

8. Crossing Classifier: Accuracy, Precision and Recall

The following metrics were reproduced from the supplied sequential dataset and immutable classifier using the original transition-level validation split (seed 42) and a 0.50 decision threshold.

84.47%	86.59%	93.09%	89.72%
Accuracy	Precision	Recall	F1 score

Metric	Value	Operational meaning
Balanced accuracy	77.25%	Average of sensitivity and specificity; useful with 72.80% positive prevalence
ROC-AUC	92.03%	Strong threshold-independent ranking of crossing vs non-crossing windows
Average precision	97.07%	Precision-recall summary across thresholds
Positive prevalence	72.80%	1,852 of 2,544 validation windows contain a horizon crossing

Confusion matrix (held-out windows)

Actual / predicted	No crossing	Crossing	Total
No crossing	TN = 425	FP = 267	692
Crossing	FN = 128	TP = 1,724	1,852
Total	553	1,991	2,544

Risk interpretation: high recall is appropriate for early-warning support because missed crossings are costly. The false-positive count shows why the probability, trajectory, confidence, and operator judgment must be presented together rather than converting the model into an automatic control action.

9. Forecast Error by Horizon

Future step	MAE	Future step	MAE
1	0.3301	7	1.5799
2	0.5708	8	1.7131
3	0.8062	9	1.8389
4	1.0247	10	1.9412
5	1.2447	11	2.0601
6	1.4231	12	2.1262

Error increases with forecast distance, from 0.3301 at step 1 to 2.1262 at step 12. GradeSense therefore returns empirical residual bands and reduces confidence as interval width grows relative to target Basis Weight.

Specification and stabilization definitions

- Specification band: target Basis Weight $\pm 2.5\%$.
- Crossing: first median forecast point outside that band.
- Remaining safe time: sample interval multiplied by first crossing step.
- Stabilization: first point beginning three consecutive forecasts within $\pm 1\%$.

What is not claimed

The residual envelope is empirical and is not a formal probabilistic calibration guarantee. The model has not been validated against a named production mill, grade portfolio, QCS calibration regime, or control strategy.

10. Recommendation Engine

For each stored forecast, the engine enumerates candidate changes across stock flow, filler flow, steam pressure, machine speed, dryer temperature, and reel tension. Each candidate must pass range, equipment, grade, rate-of-change, and dependency constraints before it can be reforecast.

Stage	Decision rule
Generate	Construct targeted single- and multi-variable setpoint candidates
Validate	Reject candidates violating equipment or grade-specific recipe rules
Simulate	Run the candidate through the same active forecast artifact
Measure	Compare crossing probability, peak deviation, and stabilization
Rank	60% risk improvement + 30% deviation improvement + 10% stabilization improvement
Filter	Omit unsafe or non-improving candidates
Present	Return action, rationale, evidence, uncertainty, and constraint attribution

Operator lifecycle

Accept -> modify/delay/apply -> observe aligned future samples -> evaluate outcome. Reject and expire paths are also persisted. This lifecycle supports recommendation accuracy and avoided/delayed crossing analysis without pretending the platform performed autonomous control.

11. Explainable AI and Inference Sources

Local explanations are computed by replacing one input at a time with its learned baseline and measuring the change in predicted off-spec probability. This yields a model-specific sensitivity ranking for the current sample.

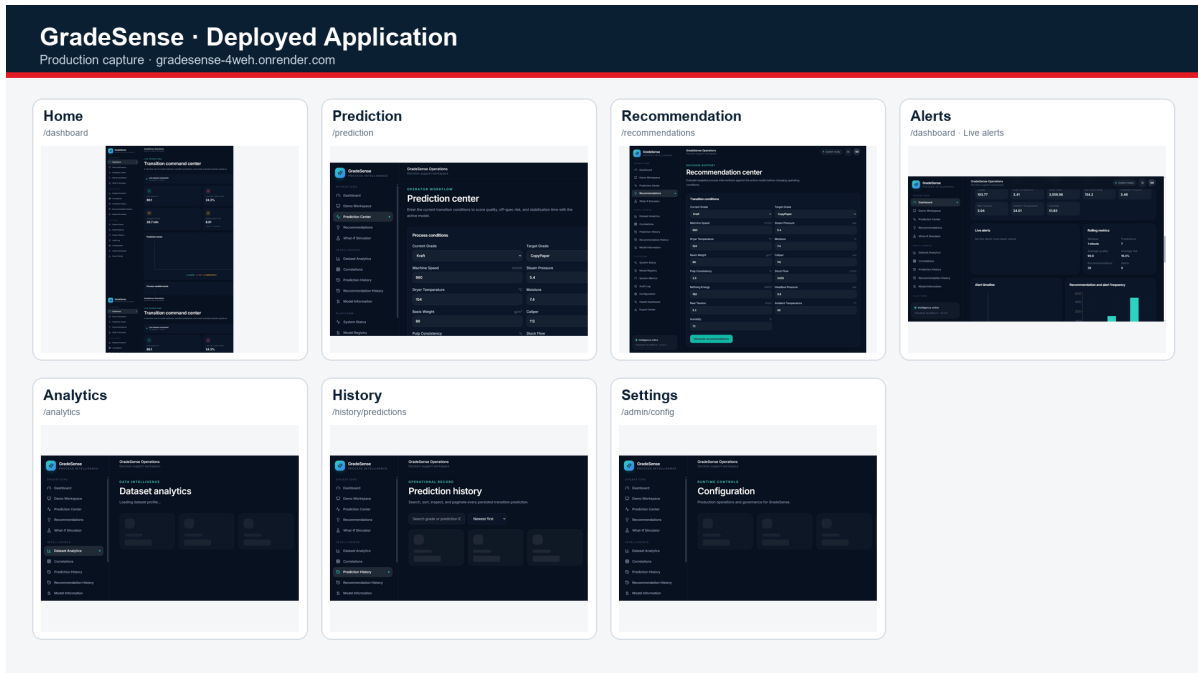
Evidence type	Use in the operator explanation
Forecast	Future trajectory, crossing risk, safe time, uncertainty
Historical trend	Direction and recent behavior of process variables
Correlation	Dataset relationships relevant to the current transition
Recipe constraint	Named rule that permits or blocks a candidate action
Similar transitions	Acceptance and evaluated outcome evidence for overlapping variables

Explainability boundary: feature sensitivity and correlation do not establish causality. A variable can be predictive without being a safe manipulated variable. Recommendation constraints and operator review are separate safety layers.

Auditability

- Model version and inference source tags travel with predictions and recommendations.
- Operator decisions and outcome evaluations are persisted.
- Registry and audit endpoints expose version, promotion, archive, and operational history.

12. Dashboard and Operator Experience



The deployed workspace covers live operations, typed prediction, constraint-aware recommendations, focused alerts, dataset analytics, prediction history, and runtime configuration.

Core pages

Home | Prediction | Recommendations | Alerts | Analytics | History | Settings. The production capture set is embedded in the presentation and retained in the submission workspace.

13. API, Persistence and WebSocket Design

Interface group	Representative operations
Prediction	POST /predict; GET /predictions; GET /model/info
Forecasting	POST /forecast; history; simulation; relationships
Recommendations	POST /recommend; intervention generation; decisions; outcomes; effectiveness
Operations	alerts, feedback, live/rolling metrics, stream status
Governance	model register/promote/archive; audit; configuration; exports; health
Real time	WS /ws/live and /ws/dashboard

Persistence model

Relational records cover predictions, recommendations, decisions, outcomes, alerts, feedback, model registry, audit, streaming sessions, and rolling metric snapshots. Model-specific trajectories, explanations, constraints, and metrics are stored as structured JSON where their schema is naturally nested.

Validation and failure behavior

- Pydantic rejects unsafe or structurally invalid inputs with HTTP 422.
- Missing artifacts fail startup explicitly rather than silently training a new model.
- Registry promotion fails if checksum, version, metrics, schema, or interfaces are incompatible.

14. Deployment Architecture

Component	Render deployment responsibility
React frontend	Static site build; client routes rewritten to index.html
FastAPI backend	Python service; Alembic migration; Uvicorn on assigned PORT
Database	External pooled PostgreSQL connection such as Neon; SSL required
Artifacts	Checked-in pre-trained models loaded read-only
Health	/health endpoint used by Render checks
Configuration	DATABASE_URL, FRONTEND_URL, VITE_API_URL and security/runtime settings

Production URL

<https://gradesense-4web.onrender.com/>

Security and scale boundary

- No built-in authentication/RBAC; deploy behind an identity-aware gateway.
- The current in-process WebSocket broadcaster and metrics buffer target one API replica.
- Multi-replica production requires a shared broker, shared rate limiting, and external metrics storage.
- Recommendations remain advisory; controlled DCS/PLC integration requires site engineering and approval.

15. Results and Business Impact

Result	Evidence	Business relevance
High snapshot fit	R2 0.9518-0.9622 across three targets	Supports useful transition-state estimates in the synthetic validation domain
Strong crossing ranking	ROC-AUC 0.9203; AP 0.9707	Prioritizes windows requiring operator attention
High crossing sensitivity	Recall 93.09%	Reduces missed early-warning events at threshold 0.50
Traceable guidance	Constraint attribution, explanations, decisions	Supports operator trust and post-event review
Deployable workflow	React + FastAPI + PostgreSQL + Render	Demonstrates an end-to-end operational product, not only a notebook

Business value is directional. The project does not claim a validated percentage reduction in waste, stabilization time, or cost. Those outcomes require a prospective mill trial with agreed baselines, grade stratification, operator protocols, and statistically sufficient transition counts.

Recommended acceptance KPIs

- Crossing recall and false alarms per transition, stratified by grade pair.
- Basis Weight MAE and interval coverage by horizon.
- Median stabilization-time reduction versus matched historical transitions.
- Off-spec tonnage avoided, recommendation acceptance rate, and evaluated recommendation accuracy.

16. Limitations, Risks and Controls

Limitation / risk	Control or next validation step
Synthetic data	Back-test on reconciled historian/QCS data; maintain transition-grouped splits
Model sensitivity is not causal	Require process-engineer review and constraint validation
Empirical intervals	Measure interval coverage and recalibrate by grade family
Class imbalance / threshold trade-off	Select threshold from site cost function; monitor precision and recall
Data drift	Monitor process/prediction drift and retrain only through governed offline workflow
Single-replica streaming	Adopt Redis/Kafka/NATS and centralized metrics for horizontal scale
No integrated identity	Place behind enterprise SSO/API gateway; add role-based authorization
Advisory recommendation	Formal management-of-change before any controlled DCS integration

Metric caveats

Accuracy can be misleading when event prevalence changes. Precision and recall depend on the 0.50 threshold. R2 does not measure calibration. ROC-AUC does not specify the selected operating point. All metrics should be tracked by grade pair, operating regime, sensor health, and time.

17. Future Scope

- Historian/OPC ingestion with time alignment, data-quality scoring, and sensor reconciliation.
- Mill-specific recipe limits, manipulated-variable eligibility, and rate-of-change commissioning.
- Threshold optimization using asymmetric costs for missed crossings and false alarms.
- Probability and interval calibration by grade family and operating regime.
- Distributed event streaming, centralized observability, and multi-replica deployment.
- Enterprise identity, authorization, electronic approval, and formal model governance.
- Prospective controlled trial measuring stabilization time, off-spec tonnage, and operator adoption.

Production validation plan

Phase	Gate
Offline	Historian replay; leakage-safe split; KPI and threshold review
Shadow	Live predictions without operational influence; drift and latency monitoring
Advisory pilot	Limited grades and trained operators; decision/outcome capture
Scale	Approved models, commissioned constraints, identity, broker, monitoring, rollback

18. References and Reproducibility

Project resources

GitHub Repository - <https://github.com/bhavukm007/GradeSense>

Live Deployment - <https://gradesense-4weh.onrender.com/>

Demo Video - [Google Drive](#)

Repository evidence used

- README.md - product scope, challenge mapping, stack, deployment and limitations.
- docs/MODEL_OVERVIEW.md - model families, validation, uncertainty and limitations.
- docs/SYSTEM_ARCHITECTURE.md - service, streaming, persistence and deployment design.
- docs/API_REFERENCE.md and docs/DEPLOYMENT.md - interfaces and operations.
- Stored joblib artifacts - versioned metrics, model objects, residual intervals and provenance.

Reproducible classifier evaluation

The report reconstructs sequential windows from the supplied CSV using history=20, horizon=12 and stride=4; shuffles unique transition IDs with NumPy default_rng seed 42; assigns 80% of transitions to training and 20% to validation; and evaluates the stored crossing classifier at threshold 0.50. No model was retrained.

Metric formulas

Accuracy = $(TP + TN) / N$ | Precision = $TP / (TP + FP)$ | Recall = $TP / (TP + FN)$ | F1 = $2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$ | MAE = mean absolute error.